Preliminary Test Setup

Barebones Test with Executables
- Upload code to Raspberry Pi Pico
- Find the Pico's serial port
- Run the windows application

Compiling the code
- Compiling for the Pico
- Compiling for the windows library

> These instructions are for a windows system.

Barebones Test with Executables

Upload code to Raspberry Pi Pico

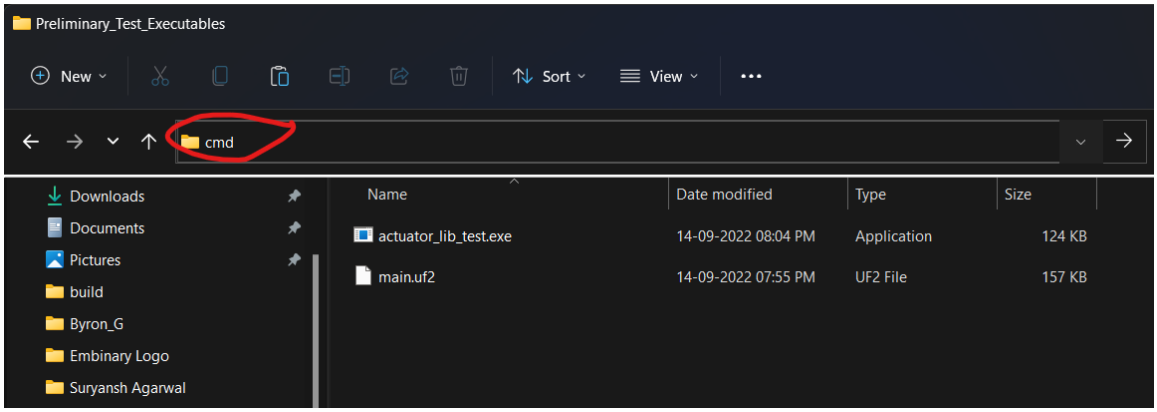
1. While holding the `BOOTSEL` button on the Pico plug it into your computer.
2. Copy the `main.uf2` file from the `Preliminary_Test_Executables` folder into the removable storage that got mounted when the Pico was plugged in.
3. After the file has been copied, the folder should automatically unmount, and the Pico should reset itself.

Find the Pico's serial port

1. Search for and open up `Device Manager`.
2. Expand `Ports (COM & LPT)` and look for a device labeled `USB Serial Device` or similar.
3. Note the COM port number. This was `COM12` in my case.

Run the windows application

1. Open the command prompt in the `Preliminary_Test_Executables` folder.
    a. This can be done by typing `cmd` in the address bar.



2. In the command prompt run the following command by replacing COM12 with your port number.

```
actuator_lib_test.exe COM12
```

3. If everything is connected successfully, this should rotate the stepper motor one way and then the other way.

Compiling the code

Compiling for the Pico

I have setup the development environment for the Pico in `Windows Subsystem for Linux`. The instructions for which are available here:

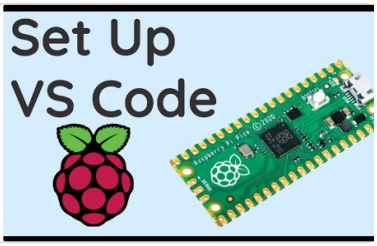
https://datasheets.raspberrypi.com/pico/getting-started-with-pico.pdf

This environment can be setup in windows natively, but the process is slightly more complicated. The instructions for this can be found here:

How to Set Up Visual Studio Code to Program the Pi Pico (Windows)

This video covers how to set up a windows machine with the required toolchain to begin programming the brand-new Raspberry Pi Pico. The toolchain has been installed and Visual Studio Code has been set up to write

https://www.youtube.com/watch?v=mUF9xjDtFfY



1. Copy the `usb-actuator` folder besides the `pico-sdk` folder.
2. Follow the same instructions as the blink program to compile the code.

Compiling for the windows library

A demo `CMake` project is provided within the `usb_actuator_lib` folder. Within this folder the `actuator_lib` folder contains the created library necessary to communicate with the microcontroller.

Requirements for compiling the provided code:

1. CMake
2. Visual Studio Build Tools

Compilation Instructions:

1. Create a `build` folder within the `usb_actuator_lib` folder
2. Open a command prompt inside the `build` folder
3. Run the following commands

```
cmake ..
maje -j4
```

4. Once the commands finish running, a `actuator_lib_test.exe` file should be created within the `Debug` folder.
5. Follow the instructions under Open the command prompt in the Preliminary_Test_Executables folder. to run the code.